ACADEMIC CENTRAL

PRACTICE EXAMINATION

# GRADE 8

# TECHNOLOGY

## PAPER 1 · PRACTICE C · MARKING GUIDELINE

**MARKS: 75**

**Marking guideline for the Academic Central Grade 8 Technology paper 1 · practice c.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

## QUESTION 1

**1** 1.1 B ✓ 1.2 D ✓ 1.3 A ✓ 1.4 C ✓ 1.5 C ✓ 1.6 A ✓ 1.7 D ✓ 1.8 B ✓ 1.9 D ✓ 1.10 B ✓ [10]

## QUESTION 2

**2.1.1** Any suitable type with a reason, e.g. a truss bridge – strong and light, can span 12 m with simple materials ✓✓; a suspension footbridge – needs few supports in the stream; a beam bridge with a central pillar (2)

**2.1.2** Any TWO with reasons: steel – strong ✓; treated wood – cheap and easy to work ✓; concrete for the supports – strong in compression and resists water; steel cable – strong in tension (2)

**2.1.3** Any TWO: compression ✓; tension ✓; bending; shear; torsion (from wind) (2)

**2.1.4** Any TWO: deep, strong foundations ✓; anchor the bridge firmly to the banks ✓; use cross-bracing; build it high above the flood level; use a wide base (2)

**2.1.5** Any TWO: handrails ✓; non-slip deck surface ✓; side netting or barriers; a maximum load sign; good lighting (2)

**2.2** A strut is in compression (it is pushed or squashed) ✓; a tie is in tension (it is pulled or stretched) ✓. (2)

**2.3** Neat sketch ✓; deck shown and labelled ✓; supports shown and labelled ✓; framework made of triangles ✓ (4)

**2.4** The material can bend, crack, buckle or break ✓ and the structure can collapse (fail) ✓. (2)

**[18]**

## QUESTION 3

**3.1** (a) Cool-drink and water bottles ✓ (b) Milk bottles, crates, shopping bags ✓ (c) Water pipes, electrical insulation, raincoats ✓ (d) Takeaway containers, packaging, cups ✓ (4)

**3.2** They do not melt or soften when they get hot ✓ and they are good electrical (and heat) insulators ✓. (2)

**3.3** To encourage people to use fewer plastic bags (reuse bags) and so reduce plastic waste ✓ (1)

**3.4** Any TWO: use refillable water bottles ✓; bring lunch in reusable containers ✓; recycle plastic in separate bins; avoid single-use straws and bags; reuse plastic for projects (2)

**3.5** A plastic made from plant materials (e.g. maize starch) that can break down more easily ✓ (1)

**3.6** A mixture of a metal with one or more other elements to improve its properties ✓. Any ONE: steel, brass, bronze, stainless steel, solder ✓ (2)

**[12]**

## QUESTION 4

**4.1.1** $48 \div 16 \checkmark = 3 \text{ times} \checkmark$ (2)

**4.1.2** $60 \times 3 \checkmark = 180 \text{ r/min} \checkmark$ (2)

**4.1.3** $\text{Distance} = 180 \times 2 \text{ m} = 360 \text{ m per minute} \checkmark; 360 \text{ m} \times 60 = 21\,600 \text{ m per hour} \checkmark = 21{,}6 \text{ km/h} \checkmark$ (3)

**4.1.4** A high gear gives more speed for each turn of the pedals ✓; on a flat road little force is needed, so the cyclist can go faster ✓. (2)

**4.2** The driver gear is turned by the input (motor or hand) ✓; the driven gear is turned by the driver and gives the output ✓. (2)

**4.3** $\text{Effort} = 600 \div 3 \checkmark = 200 \text{ N} \checkmark$ (2)

**4.4** Belt drive, any ONE: quiet, cheap, absorbs shocks, can slip to protect the machine ✓. Gear drive, any ONE: no slipping, transmits large forces, compact ✓ (2)

**4.5** (a) Open belt with both pulleys turning in the same direction ✓ (b) Crossed belt with the pulleys turning in opposite directions ✓ (2)

**4.6** A turning force (force that causes rotation) ✓ (1)

**[18]**

## QUESTION 5

**5.1.1** AND logic ✓ (1)

| Timer | Soil dry | Pump |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**5.1.2** All four input combinations ✓; output column correct ✓✓✓ (4)

**5.1.3** Correct battery symbol ✓; two switches in series ✓; motor symbol (circle with M) ✓; complete closed loop ✓ (4)

**5.2** Any TWO with what they detect: LDR – light ✓; thermistor – temperature ✓; push switch / pressure pad – pressure; microphone – sound; moisture sensor – water (2)

**5.3** (a) The bulbs become dimmer ✓ (b) The bulbs stay (about) as bright as one bulb ✓ (2)

**5.4** $\text{Saved power} = 60 - 9 = 51 \text{ W} \checkmark; \text{energy} = 51 \text{ W} \times 5 \text{ h} = 255 \text{ Wh} \checkmark = 0{,}255 \text{ kWh} \checkmark$ (3)

**5.5** An electric motor ✓ (1)

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**[17]**

**TOTAL: 75**

**END OF MARKING GUIDELINE**